

GUILHERME AMARAL

Austin, TX 78705

☎ 347-733-7687

✉ guim.amaral2004@gmail.com

🌐 linkedin.com/in/guimamaral/

🐙 github.com/guimamaral

🌐 gamaral.com

Education

The University of Texas at Austin

B.S. in Computer Science

Aug. 2022 – May 2026

Austin, TX

Technical Skills

Languages / Frameworks: C/C++, Python, Java, Clojure, Simply Scheme, AArch64, RISC-V, Bash

Technologies / Tools: Unix/Linux, Make, Valgrind, Windows, Jira, Splunk, Git, GitHub, Gerrit, Coverity, Jenkins

Experience

NVIDIA Corporation

May 2025 – Aug. 2025

Systems Software Engineer Intern – RTOS and Virtualization

Santa Clara, CA

- Expanded a legacy RTOS and virtualization memory-profiling system to support additional deployments on DRIVE Thor and DRIVE Orin SoCs across RISC-V and ARM clusters, tripling supported configurations.
- Collaborated with cross-functional stakeholders to design and deliver safety, security, and performance-focused proof-of-concepts that influenced product safety compliance.
- Developed high-reliability, performance-optimized, safety-certified C++14 APIs for MMU data capture across ARM TrustZone boundaries using AArch64 and ARM FuSa libraries.

NVIDIA Corporation

May 2024 – Aug. 2024

Systems Software Engineer Intern – RTOS and Virtualization

Santa Clara, CA

- Engineered hypervisor-kernel memory-profiling tools to visualize RAM utilization across TrustZone layers, enabling developers to better use available memory on automotive platforms (2x improvement).
- Built an ARMv8 GICv3 interrupt-profiling tool to analyze SGI/PPI/SPI behavior across TrustZone boundaries, helping SoC teams isolate hardware/software defects and bottlenecks.
- Delivered high-performance C++14 code with 100% compliance to AUTOSAR, ISO 26262, CERT, and MISRA standards, validated via Gerrit, Splunk, and Jenkins CI/CD pipelines across production, debug, and safety builds.

Technical Projects

Shellcode Encoder | *RISC-V, C*

- Engineered a UTF-8-compliant, non-NULL two-stage attack payload in assembly using a custom XOR-based shellcode encoder for 64-bit RISC-V targets.
- Optimized payload encodings to satisfy exploit constraints while reducing final binary size by 10% via bit-level transformations.

QuickJS Interpreter Runtime Exploit | *JavaScript, C*

- Exploited a use-after-free vulnerability in QuickJS to trigger type confusion and construct GC-resistant primitives: `addrof()`, `fakeobj()`, and `fakestr()`.
- Built `readmem()` and `writemem()` primitives to inspect PLT/GOT for reliable ASLR bypass, then marked pages RWX via `mprotect()` for second-stage payload execution.

Ext2 Filesystem | *x86, C++*

- Developed a full Ext2 filesystem in C++ for emulated x86 hardware on QEMU, supporting read/write operations for directories, files, and symbolic links up to 256 MiB.
- Designed a virtual memory subsystem with paging, fault handling, and file caching for kernel/user-space programs, reducing average memory access time by 80x.

Personal Website | *Next.js 14, TypeScript, React, Nginx, PM2, AWS EC2, Cloudflare*

- Built and deployed a production personal website using **Next.js 14 (App Router)** and TypeScript with a modular architecture (*site shell*, *viewport*, *content registry*) to keep navigation and content logic maintainable as pages scaled.
- Operationalized hosting on **AWS EC2** with PM2 process management, Nginx reverse proxying, TLS via Certbot, and Cloudflare edge proxying to improve reliability and secure edge-to-origin traffic.

Interests

Technical: Embedded Systems, Computer Architecture, Operating Systems, Energy-Efficient Computing, High-Performance Computing, Low-Latency, Distributed Systems, Computer Security, Formal Methods, AI Safety